James Forward

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EDUCATION

Incoming Ph.D. Student, Systems Biology 09.2026 – *Present*
Harvard Medical School, *Cambridge, MA*
Herchel Smith Graduate Fellow

B.Sc. Integrated Science (Microbiology and Human Biology), with Distinction 09.2019 – 05.2024
Faculty of Science, University of British Columbia (UBC), *Vancouver, BC, Canada*

RESEARCH EXPERIENCE

University of California, San Francisco (UCSF), *San Francisco, CA* 09.2024 – *Present*
Post-Baccalaureate Researcher, Cell and Tissue Biology
Advisors: Fred Chang and John Vaughen

- **Project 1**: Leading a project on how signaling and developmental programs trigger global changes in the physical properties of the cytoplasm.
  - Using single particle tracking of Genetically Encoded Multimeric nanoparticles (GEMs) and TIRF microscopy, I am using the pheromone signaling pathway in *Saccharomyces cerevisiae* as a model to probe how the cytoplasm organizes its components as the cells differentiate for mating.
  - Initiated a collaboration with the Manalis lab at MIT to perform single cell intracellular density measurements using a suspended microchannel resonator (SMR)
  - This work has led to one first author manuscript (in preparation), two invited talks, and two poster presentations.

- **Project 2**: Spearheaded a project that aims to understand how principles of cytoplasmic crowding change at the multicellular level in *Drosophila* brains.
  - Established that GEMs can be robustly expressed and tracked using confocal microscopy in *ex vivo* brain preps with cell-type and sub-cellular resolution—one of the first demonstrations of this tool in a live tissue.
  - Developed a custom pipeline to extract diffusion coefficients that excludes deconvolution artifacts and aggregates.
  - Initiated a collaboration with the O'Brien Lab at Stanford to image GEMs in *Drosophila* gut cells.
  - This work has led to one first author manuscript (in preparation), two invited talks, and three poster presentations.

Marine Biological Laboratory (MBL), *Woods Hole, MA* 06.2025 – 07.2025
Whitman Center Assistant
Advisor: Fred Chang

- Using nanorheology and phase microscopy on *Saccharomyces cerevisiae* to understand how intracellular density and mechanics of the cytoplasm are regulated during rapid changes in temperature.
- Attended talks from several MBL courses: Physiology, Neural Systems and Behavior, Microbial Ecology

UBC Zoology & Microbiology & Immunology, *Vancouver, BC* 02.2023 – 09.2024
Award Funded Undergraduate Thesis Student
Advisor: Kayla King

- Conceived a project for my undergraduate thesis on how temperature affects bacterial morphogenesis and growth over evolutionary time.
- Using a combination of single-cell imaging and computational modelling, I analyzed image stacks for changes in cell size in 'Morphometrics' on MATLAB.
- Quantified both single cell and population level growth measurements to associate with levels of fitness or virulence.
- In parallel, I successfully developed methods to engineer the lab's undomesticated non-model bacterium, *Leucobacter* sp., through a transposon-based plasmid to enable visualization and tracking of competition between evolved and ancestor populations
- This work has led to one invited talk, two poster presentations, and one co-author manuscript that has been provisionally accepted at *Proceedings of the National Academy of Sciences* (PNAS)

UBC Microbiology & Immunology, *Vancouver, BC* 01.2024 – 08.2024
Undergraduate Researcher, *Course Based Research*
Advisor: Evelyn Sun

- Led a project that investigated the role of inflammation and surgical intervention on the gut microbiome in Crohn's Disease.
- Utilizing a dataset with 16S ribosomal RNA sequencing data and fecal calprotectin measurements, we examined how inflammation and bowel resections affect the microbiome's composition and diversity.
- This work had led to one invited talk and one co-first author publication.

UBC Microbiology & Immunology, *Vancouver, BC* 09.2022 – 04.2023
Undergraduate Researcher
Advisor: Brett Finlay

- Assessed *E. coli* Nissle's probiotic qualities by studying the operational dynamics of its Type Six Secretion System (T6SS).
- Employed protein analysis suites to analyze domain function and the specificity of effector proteins to the secretion system.
- Determined virulence factor and gene expression in response to environmental stimuli via reporter gene assays.
- This work has led to one co-author manuscript that is to be submitted to *Cell Host & Microbe*

LEADERSHIP EXPERIENCE

Co-President, UBC Neuroscience Society (Now UBC Neuroscience Association) 04.2023 – 05.2024
- Led a team of 30 Execs to run an academic society of 500+ undergraduate students, which also partners with the UBC Neuroscience Department.
- Spearheaded the annual UBC Neuroscience Undergraduate Research Conference (>400 attendees). We gave undergraduate students a platform to present their research in talks and posters and included keynote speeches from faculty members and sponsored workshops.
- Wrote grant applications and was awarded $18k+ for use in the 2023/24 academic year

Science Undergraduate Research Mentorship Program, UBC Faculty of Science 09.2022 – *Present*

- Provided mentorship and leadership to 10 undergraduate students based on their goals, notably helping several students secure funded research experience in academic labs.

Neuroscience Steering Committee, UBC Department of Neuroscience 04.2023 – 05.2024

- Worked closely with the UBC Neuroscience department and contributed to the decision-making process that shape the direction and initiatives of the department.
- Collaborated with course developers and the director of the neuroscience program to design the undergraduate course: Neuroscience Capstone (NSCI 400)

VP External, UBC Neuroscience Society 04.2022 – 05.2023

- Oversaw a team of 8 students to organize external events for 600+ undergraduate attendees throughout the year that focus on neuroscience-related careers, networking for students, and community building.

Instructor, AllSet learning 05.2019 – 08.2023

- Collaborated with other coworkers to create lecture slides and problem sets for high school science and math, AP biology, and AP chemistry.
- Filmed work examples to teach the students during the Covid lockdown.

PUBLICATIONS

Journal Articles

1. Vancomycin promotes key microbiota-pathogen interactions and removes protective bottlenecks to Citrobacter rodentium infection
   Woodward SE, Peña-Díaz J, Serapio-Palacios A, Vogt SL, Wang MA, Feng W, **Forward J**, Huus KE, Krekhno Z, Neufeld LMP, Cirstea M, Lee A, Finlay BB.
   To be submitted to *Cell Host & Microbe*

2. Rising temperatures favour parasite virulence and parallel molecular evolution following a host jump
   Hector TH, Kreiner J, **Forward J**, Hoang K, Stevens E, Johnson S, Li J, King KC.
   Provisionally accepted at *Proceedings of the National Academy of Sciences (PNAS)*
   bioRxiv DOI: 10.1101/2025.01.14.632940

3. Surgical intervention correlates with reduced bacterial diversity in the gut microbiome of Crohn's Disease patients exhibiting low levels of inflammation.
   Bremner M, Feng A, **Forward J**, Guan R, Zhang S. (2024).
   Undergraduate Journal of Experimental Microbiology, Peer Reviewed (UJEMI+)
   a. All authors contributed equally to project conception, experimentation, and writing

AWARDS AND SCHOLARSHIPS

Herchel Smith Graduate Fellowship at Harvard and Cambridge University 03.2026
Shurl and Kay Curci Foundation Scholarship at UCSF – *Declined* 03.2026
Stanford Doctoral Research Assistantship, $56,348 + full tuition / yr - *Declined* 03.2026
Rockefeller Doctoral Research Assistantship, $55,620 + full tuition / yr - *Declined* 03.2026
Columbia Doctoral Research Assistantship, $48,551 + full tuition / yr - *Declined* 03.2026
Harvard Doctoral Research Assistantship, $51,500 + full tuition / yr 02.2026

UCSF Doctoral Research Assistantship, $52,211 + full tuition / yr - *Declined* 02.2026
UBC Science Undergraduate Research Experience (SURE) Award 05.2024
UBC AMS Nomination for Best Academic Club *during my presidency* 04.2024
UBC Dean's List 09.2019 - 05.2024
British Columbia Government Achievement Scholarship 09.2019
British Columbia Government Award of Excellence 09.2019
BC First Responders Community Award 09.2019

SELECTED TALKS

Dynamics of Cytoplasmic Properties: From Single Cells to Multicellular Organisms
Forward J, Wu Y, Mettinen T, Lemière J, Vaughen J, Chang F
- Bay Area Cytoskeleton Symposium, San Francisco CA 05.2026
- UCSF Tetrad Graduate Student Retreat 2025, Pacific Grove CA 09.2025
- Bay Area Cytoskeleton Symposium, San Francisco CA 04.2025

Mating Pheromone Induces Increased Cytoplasmic Diffusion In Budding Yeast
Forward J, Lemière J, Chang F
- 4th Annual Cell Size and Growth Meeting, Lake Tahoe CA 02.2025

How Growth and Form Adapt in Pathogens to Temperature Changes over Evolutionary Time
Forward J, McCallum G, Hector T, Tropini C, King K.
- Host-Parasite Supergroup Meeting, Vancouver BC 04.2024

Investigating the Role of Inflammation and Surgical Intervention on the Gut Microbiome in Crohn's Disease
Forward J, Bremner M, Feng A, Guan R, Zhang S.
- Microbiology Undergraduate Research Symposium, Vancouver BC 04.2024

POSTER PRESENTATIONS

Dynamics of Cytoplasmic Properties: From Single Cells to Multicellular Organisms
Forward J, Wu Y, Mettinen T, Lemière J, Vaughen J, Chang F
- Bay Area Cytoskeleton Symposium, San Francisco CA 05.2026
- Cell Bio 2025, ASCB/EMBO Meeting, Philadelphia PA 12.2025
- UCSF Tetrad Graduate Student Retreat 2025, Pacific Grove CA 09.2025
- Chan Zuckerburg Biohub, Physics of Life Symposium, San Francisco CA 09.2025

Mating Pheromone Induces Increased Cytoplasmic Diffusion In Budding Yeast
Forward J, Lemière J, F Chang
- Bay Area Cytoskeleton Symposium, San Francisco CA 04.2025
- Cell Bio 2024, ASCB/EMBO Meeting, San Diego CA 12.2024

How Growth and Form Adapt in Pathogens to Temperature Changes over Evolutionary Time
Forward J, McCallum G, Hector T, Tropini C, King K.
- Chan Zuckerburg Biohub, Physics of Life Symposium, San Francisco CA 09.2024
- Canadian Society of Ecology and Evolution 2024, Vancouver BC 05.2024
- UBC Multidisciplinary Undergraduate Research Conference (MURC), Vancouver BC 03.2024